MOOCs & STEM Fields

Rethinking Assessments

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Abstract. In this paper I uncover how MOOCs have changed the way that STEM educators at MIT conceptualize assessments and pedagogy. I show that blending a classroom can lead to a more careful and precise approach to problem and assessment generation as well as expand the scope of problems asked on formal assessments or homework, thus requiring a higher degree of knowledge/understanding from students when compared to a traditional classroom setting. I also analyze the *inadvertent scaffolding* sometimes induced by technological limitations when using MOOC materials in conjunction with a traditional classroom and how this shifts pedagogical paradigms.

Overview

We live in a world today where technology and innovation are driving massive cultural and educational shifts. With the advent of such a vast array of new technologies, it is clear that the content of *what* we teach, particularly in science, technology, engineering, and mathematics (STEM) fields must change. Recently, however, such technological advancements have allowed for a shift in *how* we educate. Within the past five years massive open online courses (MOOCs) have appeared through websites such as Coursera and Udacity (both founded by Stanford affiliates), as well as the dual Massachusetts Institute of Technology (MIT) and Harvard project edX.

While free MOOCs offered through associations such as edX have the obvious unambiguously beneficial effect of making secondary and post-secondary education more available on a global scale, especially to impoverished communities without higher educational systems, MOOCs are not without criticism. A quick Google search will bring up countless of hits of newspaper articles, op-eds, blogs, and pretty much every other sort of media criticizing MOOCs. One of the major criticisms of MOOCs lies in their modes of assessment. Without a teacher to asses the student' understanding directly, and many possible avenues for students to falsify answers, many criticize MOOCs, claiming that they cannot properly assess students' understanding of the material. Another major concern is that online assessments lend themselves to multiple choice problems, which are not able to holistically asses students' understanding of the subject matter.

This argument, however, becomes much less valid when materials from MOOCs are used in conjunction with a traditional classroom, creating a so-called *blended classroom*, in the sense of Alonso et al. (2005:232–233). In blended classrooms, a teacher or professor is still present, but online materials provide a supplementary classroom experience for the students. In this paper I uncover how developing MOOCs has changed the way in which educators conceptualize assessments and pedagogy. I show that blending a classroom can lead to a more careful and precise approach to problem and assessment generation as well as expand the scope of problems asked on formal assessments or homework. I also describe how blended classrooms have the ability to offer *contextualized* or *adaptive scaffolding* on homework assignments through various hinting procedures (Vygotsky 1978). Lastly, I discuss how more precise problems can lead to unwanted *inadvertent scaffolding* (Vygotsky 1978), and how educators work to get around this issue.

Background

To uncover the effects of MOOCs at MIT, I interviewed three MIT MOOC developers: Drs. Jennifer "Jen" French, Saif Rayyan, and Simona Socrate. Jen is the program manager for MITx in the mathematics department and former postdoctoral associate at MIT's Teaching and Learning Laboratory (TLL). Jen mainly works on creating online materials for MIT's residential single-variable calculus and differential equations courses. Her single-variable calculus course is going to run as a MOOC on edX starting summer 2015. Saif is also an educational researcher and lecturer in the physics department. He has been part of edX since its conception and has been researching the use of technology in the classroom as well as online education for many years. Simona is also a principal research scientist and senior lecturer in the mechanical engineering department. Simona's work on MOOCs has been devoted to her popular edX course *Elements of Structures* which she also teaches residentially at MIT using the same online materials.

All three MOOC developers indicated that these online materials have a characteristic organizational flow. The material is broken down into small units such as *days* or *lectures*, and that material is broken down even further into *sections* (French 2014). Each day generally consists of the material that would be covered in one lecture in a traditional classroom setting, possibly with additional worked examples or supplementary readings (French 2014). This information is usually disseminated through a variety of means, including lecture videos broken up into shorter videos with *concept questions*, short questions with multiple attempts[1] designed to quickly asses the students' understanding of concepts discussed in the videos, in between video segments, text, and interactive visualizations or *learning sequences*. Concept questions are also dispersed throughout the text with an analogous goal.

These concept questions are one of the fundamental features of MOOCs, and may come in one of several standard flavors, or alternatively may be designed separately

[1] By this I mean that student have multiple chances to answer the question.

by the course instructor. The standard problem types available today include multiple choice, numerical response where students can enter in a number as their answer, and formula response where students enter in a mathematical formula as their answer (French 2014). Some of the more common instructor-designed problem types include Matlab[2] based problems, problems involving javascript visualization activities, and "drag and drop" problems where students can drag images or text around to the appropriate location to complete a diagram, for example (Socrate 2014). Open-ended or "free response" type problems are uncommon due to staff and technological limitations regarding grading of responses (Socrate 2014).

The Blended Classroom

Let me first specify what I mean by the term *blended classroom*. For the purposes of this paper, a traditional (lecture-based) classroom will be considered a *blended classroom* if the course uses online materials to provide a supplementary classroom experience for the students as well as to change and in-class experience. Such a classroom might have online javascript applet demonstrations in class, use some sort of visualization software available to students in class as a "lab", provide lecture videos or videos of worked examples online, or in general use any of the MOOC materials discussed in the previous section in conjunction with traditional lectures and methods of pedagogy, hence the definition I adopt here is very much in alignment with the definition of Alonso et al. (2005:232–233). The "online classroom" then becomes a sort of a metaphysical extension of the physical classroom where students can further explore course materials as well as deepen their understanding of key concepts via answering concept questions and going through worked examples.

Concept questions are one of the key features that distinguish MOOC materials, even those that consist mainly of text, from traditional textbooks as supplementary course materials and really make the online classroom a metaphysical extension of the learning community from in the physical classroom. While some texts may ask concept questions, the solutions are generally worked out below and do not lend to students answering the concept questions, but rather, teach students to simply read the solutions. On the other hand, concept questions in MOOCs do not present the solution straight-off, and hence give the students a better opportunity to think about and work through the answer themselves before they read the solution, thus extending the sort of dialog that one might find in a traditional classroom setting to the online classroom.

From the standpoint of efficiency, the use of MOOCs in traditional courses to create a blended classroom, can make a traditional course more efficient to run for the educator or team in charge of the course. All three interviewees indicated that when using a

[2] A high-level computer language and interactive environment for numerical computation, visualization, and programming.

MOOCs in conjunction with a course, they are able to put an “easy-to-grade” portion of homework assignments online, thus reducing the volume of work of graduate student teaching assistants and undergraduate graders. But moving easy-to-grade problems online has another consequence. In Jennifer French’s differential equations course, each homework assignment has two parts; the first part has historically consisted of more mechanical “book problems” that are easy to grade and simply test students’ ability to mechanically solve differential equations, while the second part is more open-ended and may require graphing, solving real-life problems, or any number of higher-order thinking tasks. Jen indicated that since she has started to grade the first part of the homework for the differential equations course she runs online, it has changed the way she has thought about what sorts of questions to ask:

> …I would say we held students accountable for conceptual understanding but didn’t asses them on it before, but things like multiple choice questions and these other sorts of other very easy to grade problem types have made us ask these kinds of questions and hold students accountable for it which I actually think is quite positive. These questions were never things we put on open-ended assignments before.

So not only did putting part of the homework online make the course more efficient (and actually allow graders to spend more time giving feedback on the second part of the homework), but it also helped the course address learning objectives that were not really given as practice, but students were expected to know for formal assessments; a clear misalignment of pedagogy (French 2014).

There is another key benefit of online homework. Online problems have the key benefit of being graded immediately. Jen indicated that not only is she asking different questions, but “they’re [the students] getting feedback on how they’re doing immediately as soon as they do it.” Jen also indicated that there is another (optional) feature that does not exist for a traditional homework assessments, regardless of whether or not the types of problems are changed:

> There’s also crowd sourced hints which I think are pretty cool which actually work in real time where people who make errors can actually give hints that helped them so if somebody else enters the same expression then they’ll get that hin…how much better is it to tell them “Hey this isn’t…right, take this opportunity to understand where you’re wrong and fix it.” I think that’s more powerful — I think that as a learning tool this is actually the ideal.

Moreover, after submission these hints are vetted by course instructors and teaching assistants who make sure the hints actually address the mistake correctly and lead students in the proper direction (Socrate 2014). Indeed, the ability to get the instantaneous feedback of whether a student answered a homework question correctly or not is a feature that does not exist in a traditional lecture-based classroom setting, but the additional crowd-sourced hinting option is even more powerful.

Not only do crowd sourced hints provide students with instantaneous feedback, but they also give the opportunity for students to be nudged in the right direction by someone who got stuck in the same spot, which is much more powerful than just telling the student that he/she is incorrect. This seems to be an unambiguous positive for questions with a definite answer. Simona also indicated that she felt that the crowd source hints were very helpful in her Elements of Structures course, both online and residentially, indicating that, after the introduction of crowd sourced hints, students were able to answer problems correctly in their allotted number of attempts more frequently (Socrate 2014).

There are reasons for this other than the fact that students are getting hints to problems. Academic communities are language communities, and students are newcomers to these communities while professors, instructors, and problem developers are old timers who have developed expert registers. A potential reason why a student may answer a problem incorrectly is because he/she does not understand the language used in the problem well enough to be able to correctly interpret what is being asked. Other students are much more likely to be within the Vygotsky's zone of proximal development (1978), and may be able to phrase questions in a way that is clearer to other students.

These online learning platforms also provide alternative means of providing hints. As Saif indicates, instructors can provide hints that guide a student through the process of solving a problem when the student does not answer a problem correctly on his/her first attempt:

> …Dave [Pritchard][3] created a tutorial to do physics problems. Basically he said you try the problem and if you cannot do it, "do you want a hint?" And the hint was presented as a structure. So he basically presented it the way that you solve the problem [asking] "how would you think about this problem?" and broke it into steps. Now each step was also interactive…and in each step you have questions so even in the hint you would answer sort of intermediate steps and I think this is what led to the learning because now we provided the students with a way instead of looking at the problem and just staring at it and having to go cheat, they have a way to get feedback and do the problem.

What both of these methods of hinting do is provide *contextualized* or *adaptive scaffolding* to problems, by which I mean, scaffolding, but only in the context that a student cannot solve a problem on his/her own and actually needs the scaffolding. Both of these modes of hinting are in a sense quite similar to the scaffolding that Downey presents where novice capoeira students "learn a great deal from imitating contemporary practice" (2008:204). In the first mode, the student is, as mentioned, being nudged in the right direction by another student within the zone of proximal development. In the second case, the mode of "contemporary practice" is a set structure for solving a specific problem designed by an expert. Both of these are exhibited in Downey's analysis of

[3] MIT physicist and educational researcher.

capoeira instruction (2002), though the role of a "senior student" might be replaced by a peer who has already completed the problem (correctly). By doing so, the instructor is able to extend the physical classroom to the online classroom creating a metaphysical classroom — it is as if the instructor were walking around class, right there able to provide the proper scaffolding for a problem if needed without the instructor being present at all. From a pedagogical point of view, this is clearly superior to students being disengaged in learning because they are unable to answer questions — this forces the student to be engaged in the learning process. Moreover, it helps further the social reproduction of the methods and ideals of the subject commonly recognized by experts in a given field (Bourdieu:78–81).

Rethinking How to Ask Questions

As I just illustrated, using MOOC materials in a classroom makes educators rethink the content of *what* sorts of questions they ask as well as how to use scaffolding to nudge students in the correct direction when they get stuck. Now I present an argument that using MOOCs in conjunction with a traditional classroom forces the astute educator to more carefully think about *how* he/she poses questions. When asked if the problems being online (and potentially available to tens of thousands of people in the case of a MOOC) forces an instructor to think a lot more about their pedagogy one, and also make problems better, Simona Socrate responded an affirmative "Yes. The short answer is yes by a lot." She also indicated that "The problems themselves need to be very well thought out otherwise it's a waste of time." Now the inevitable question arises, *Well, isn't this true in a standard lecture-based classroom?* I think that anyone who has taken a few (college-level classes) would agree that though the answer *should be* yes, this does not always happen. Simona echoed this claim. As part of a committee analyzing the curriculum of the mechanical engineering department and indicated that while analyzing homework problems from various courses she realized that "…sometimes the problems are not well-posed…or they're ambiguous in the way the statement is put together" as well as that there is often missing information (Socrate 2014).

Even relatively large lectures are quite small when compared to the scale of many MOOCs (about an order of magnitude). This being the case, it is not at all uncommon for professors to give ambiguous or incorrect assignments and then have to e-mail out to the class to correct the assignment or announce the correction in lecture. On the scale of a MOOC, this is much more difficult to do effectively because it is much more difficult to communicate with 10,000 people who you've never met from across the world than it is to communicate with 500 people you see three times a week in lecture. This forces instructors using online MOOC materials to be more precise in their problem statements. Simona indicated that:

> In general there is the tendency, when you write your own homework, to be a little, not sloppy, but not as rigorous. A lot of times the homeworks are written by graduate students who have a lot of time constraints so they try to do a good job, but I mean it takes a lot of effort to write a good problem…so I think that creating this material online forces you to be so much more careful about how you state your problems and you have to make your problems so robust. And it definitely benefited the residential students.

So in the sense of definiteness of problems, some of the technical difficulties that come from trying to formulate a problem precisely online actually result in problems that are generally more often well-posed and precise in the statement of what they want in the solution, both of which are things that often do not happen in a traditional classroom. In the vast majority of situations instructors deem this desirable as it helps insure the well-posedness of problems, which is beneficial logistically when it comes down to grading, et cetera.

Too Much Scaffolding?

The effects of online problems presented thus far are perhaps slightly romanticized. They do not come without challenges regarding creating the problems as well as limitations on the different types of problems that can be asked; open-ended questions are nearly impossible to ask, for instance (Rayyan 2014). But there is another challenge — a challenge involving scaffolding. Earlier I presented an argument that the sort of *contextualized scaffolding* that instructors are able to provide through the use of MOOC materials which they cannot normally provide on paper homework assignments, for example, is a beneficial method of pedagogy, but there is a type of scaffolding that the analysis so far has missed, something I will call *inadvertent scaffolding*. I will say that problem scaffolding is *inadvertent* if the scaffolding is introduced into the problem because of limited procedures for asking questions which result in questions posed in a manner which differs from how it would be posed in the physical classroom or on a (static) paper homework assignment. Simona told me that sometimes problems are changed to be perhaps too precise in order to be put online.

> Sometimes I wonder if we [MOOC developers] are…almost too precise in these problems. Like for instance the fact that for symbolic questions sometimes we tell them give us, you know, *q* in terms of *a*, *b*, *c*, *d*, and *f*. And I think sometimes we tend to help them too much, so I think we have to rethink some of that.

Jen and Saif also mentioned similar phenomena in my interviews with them. Simona also indicated that because initially the online grader wasn't robust enough, she had do significantly alter the way that she posed questions in order to make student responses readable by the grader.

> In the class that I have now I tell them exactly which symbolic variables to use, but there are other courses I've seen in which they say "you can enter your answer and any of these symbols may be in the answer." And so you give them the whole list of symbols in the problem and tell them how to enter it, and then it's up to them to choose which symbols to enter in a specific answer.

Hence this is a clear introduction of inadvertent scaffolding into a problem. Simona also indicated that they have also changed the paradigm for multi-part problems. Before they might have just asked for the answer to the fourth part of a multi-step procedure, for example, but now they'll walk students through all of the proceeding parts as well, which doesn't present the challenge of having to know that those parts need to be completed before the fourth part can be completed. Moreover this restructuring of problems is solely due to the limitations on the ways that online problems can be structured (Socrate 2014). Hence again this is a clear example of inadvertent scaffolding; though there are places where it is appropriate to use such scaffolding as a pedagogical too, instructors do not always want to use so much scaffolding, but they are limited by the current technologies so they introduce more scaffolding than they would like to. This increased scaffolding then has the potentially deleterious effect of students not learning, or not learning as well, how to work outside of the framework given in online problems. Jen, Saif, and Simona all indicated that they alter what happens in the physical classroom to get around these issues. Moreover, all of them have reported significantly higher learning outcomes using online materials in conjunction with the traditional classroom compared with the traditional class setup, thus suggesting that the net effect of these two types of scaffolding may be positive (French 2014).

Conclusion

Massive open online courses have changed the way that we think of education on a global scale. In this paper, I discuss how MOOCs have also fundamentally changed some of the ways that educators think about student assignments. Primarily, the use of MOOC materials in a blended classroom has changed the content of what educators ask students on homework assignments. Primarily, using educational technologies within a course gives educators the opportunity to ask questions they otherwise would not because of limited (grading) resources. Secondarily, hinting procedures allow instructors to provide contextualized or adaptive scaffolding, giving students support only when they cannot answer questions on their own. Crowd-sourced hinting procedures allow instructors to take advantage of the zone of proximal development in these instances, both in the obvious sense and in the sense of language socialization into STEM fields.

On the other hand, using online materials in traditional classrooms forces educators to shift the the paradigms they use to pose questions due to technological limitations. This often results in an inadvertent addition of scaffolding to problems which would otherwise not be apparent in traditional settings. This in turn causes educators to create new pedagogical methods for framing problems in order to insure that students are able to eventually remove themselves from the problem scaffolding. Though the net effect of these two sides of scaffolding is not clear, instructors using the blended classroom paradigm have reported higher measurable learning outcomes in classes using supplementary online materials when compared to traditional classrooms, suggesting that in this instance the potential negative effects of induced scaffolding may be counteracted by the positive effects of contextualized scaffolding.

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